

Nitrogen Vacancy Centers for Quantum Sensing

Auden Young

What is a color center?

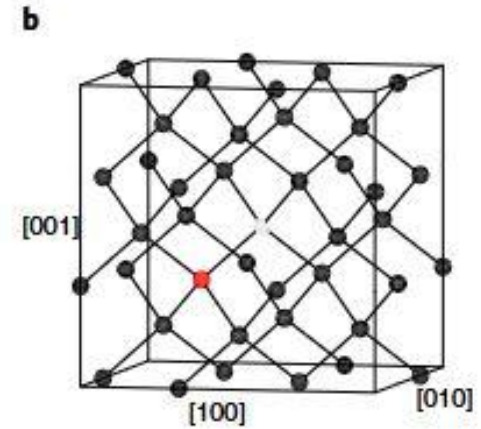
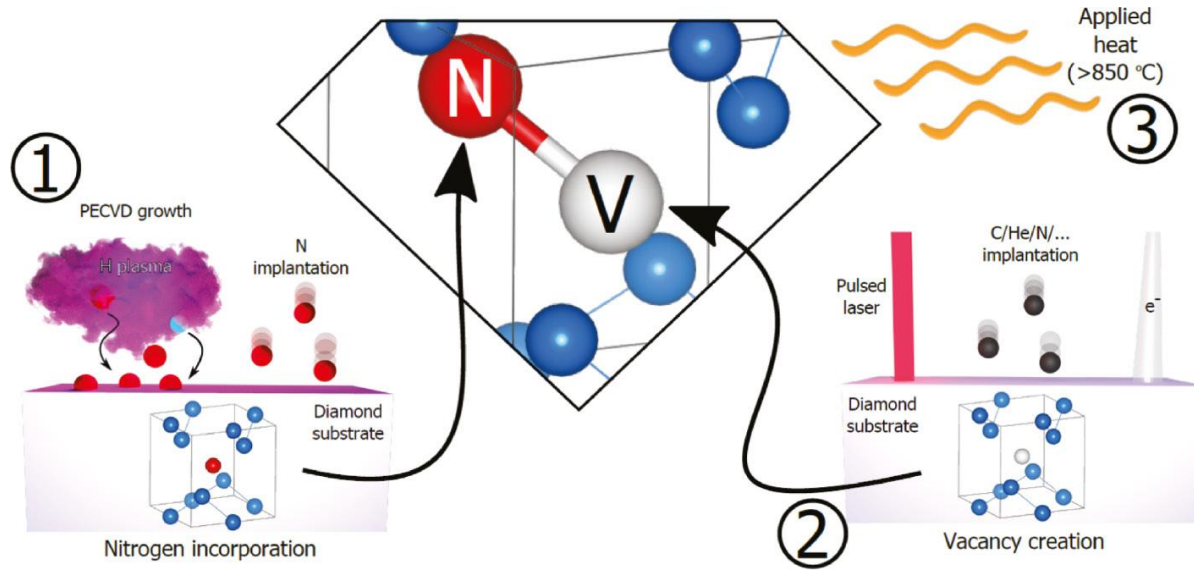


Figure from Bucher et al.

Nitrogen vacancy (NV) centers

- Initialize and readout NVC with ODMR spectroscopy
- $m_s = 0$ using 532 nm laser
- Manipulate and drive spin states between $m_s = 0$ and $m_s = \pm 1$ using microwaves
- Zero field splitting ~ 2.87 GHz at room temp; external magnetic field lifts degeneracy of $m_s = \pm 1$ via Zeeman effect, with

$$\Delta f = 2\gamma B_{NVC},$$

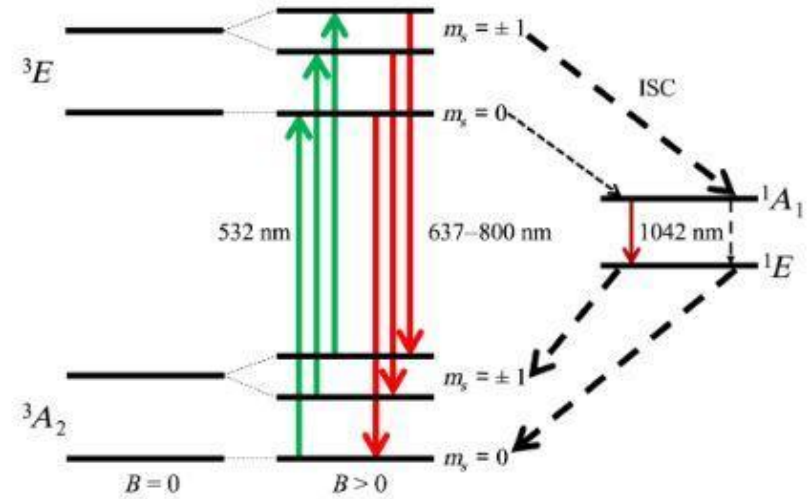


FIG. 5. A schematic of the electronic structure of the NVC in diamond, with electric dipole transitions and intersystem crossing (ISC) indicated.

Adapted from Graham et al., 2023

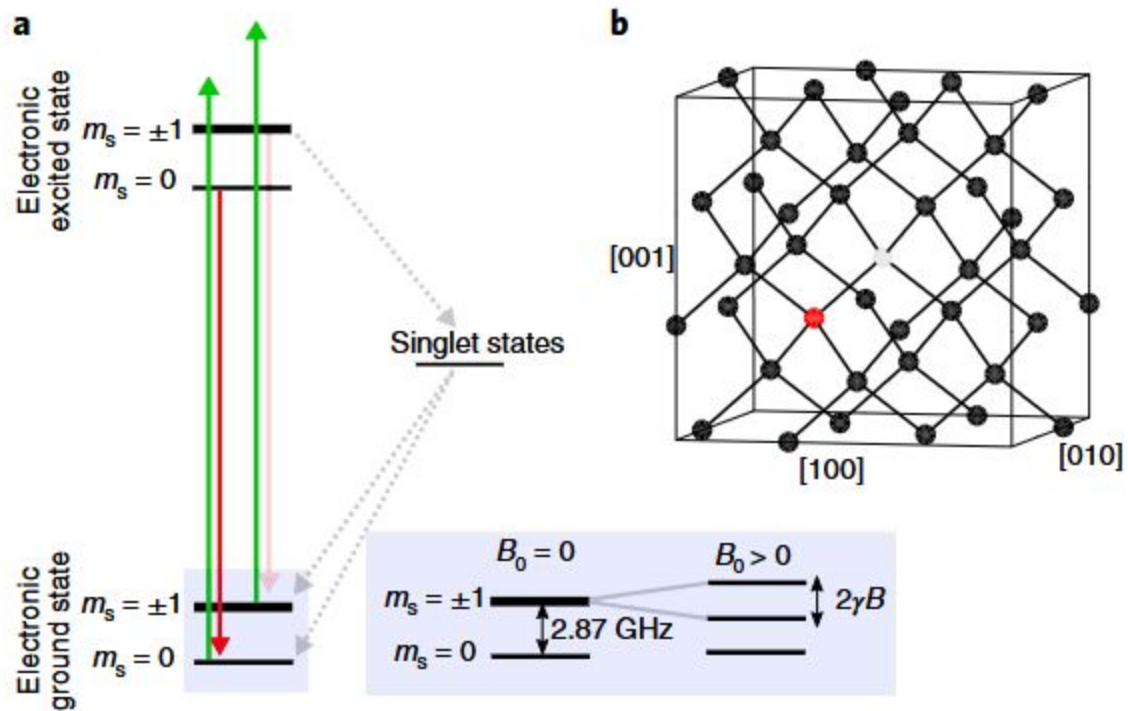
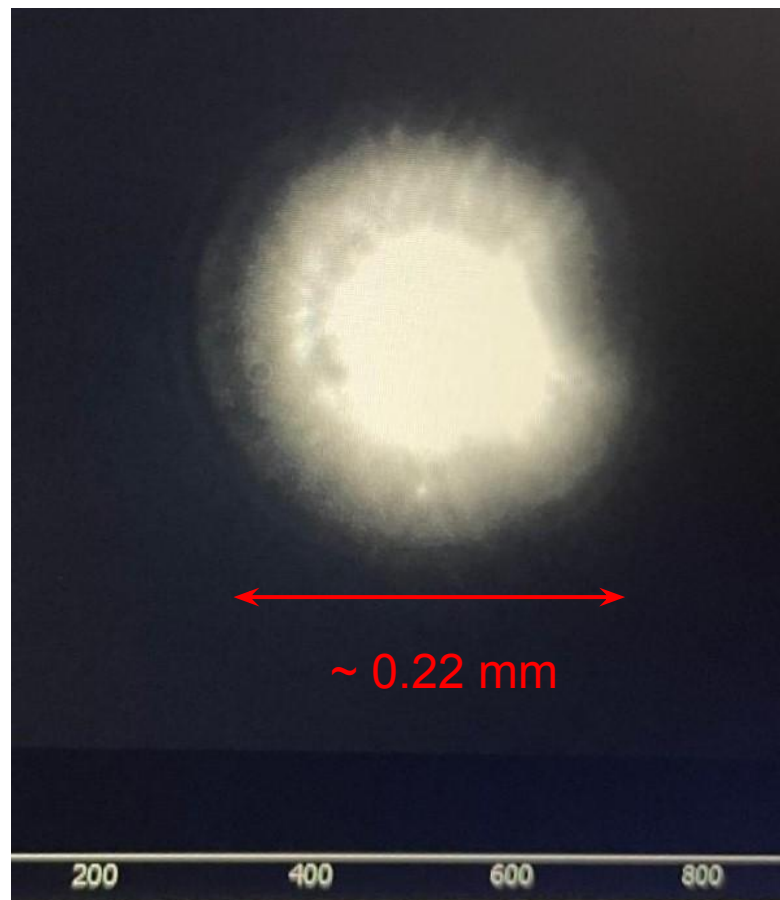


Figure from Bucher et al.



Zeeman effect

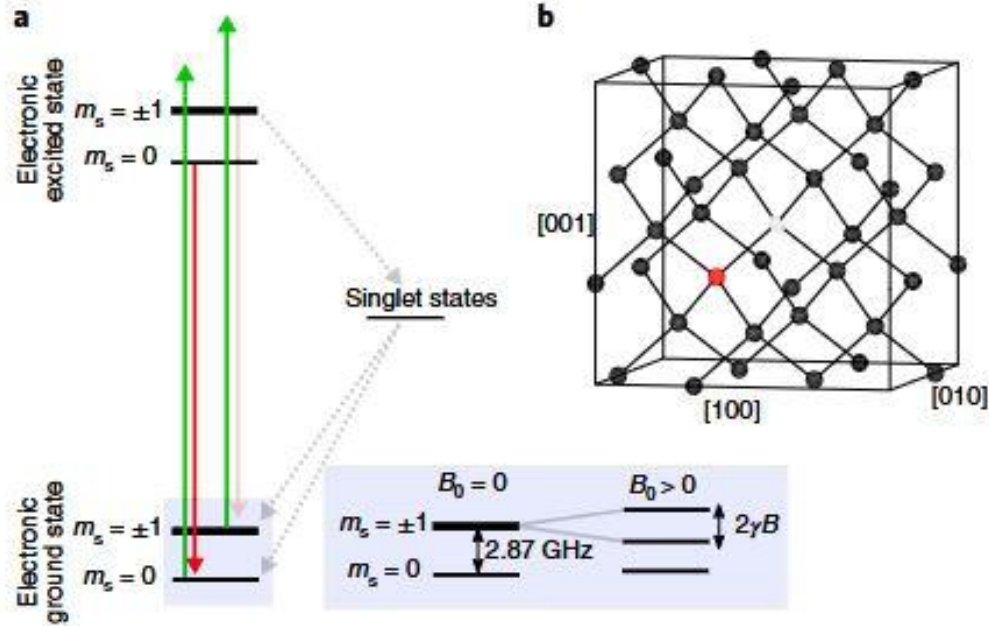
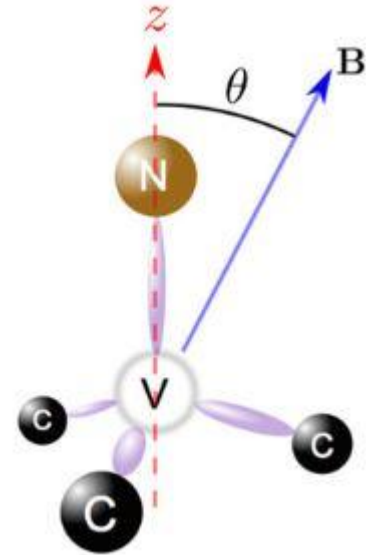


Figure from Bucher et al.



$$B_0 = \frac{2,870 \text{ MHz} - f_{\text{NV}} (\text{MHz})}{2.8 \text{ MHz/G}}$$

Controlling NV centers

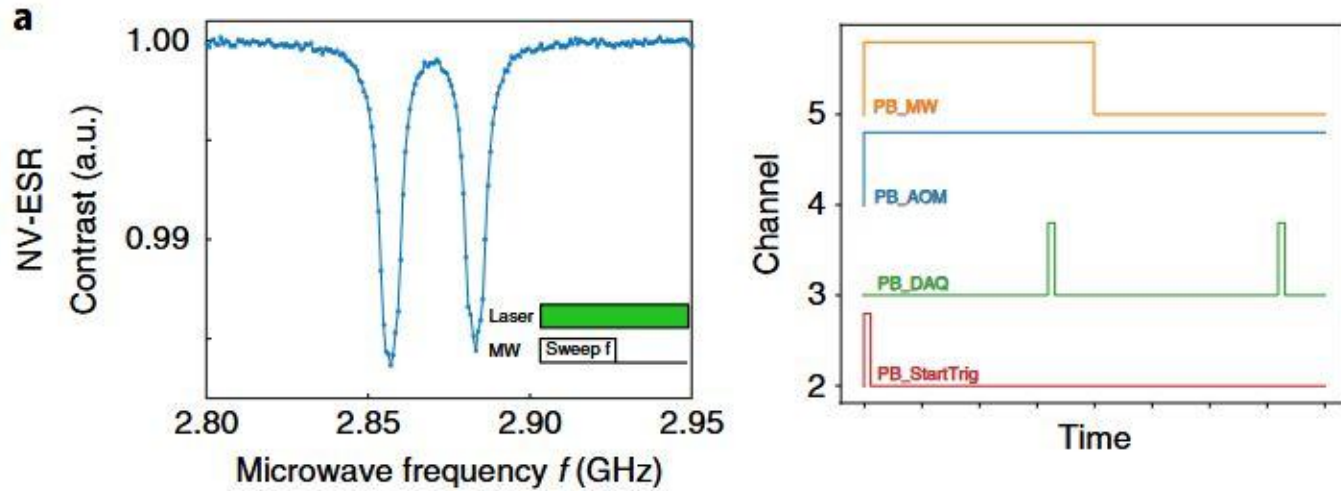


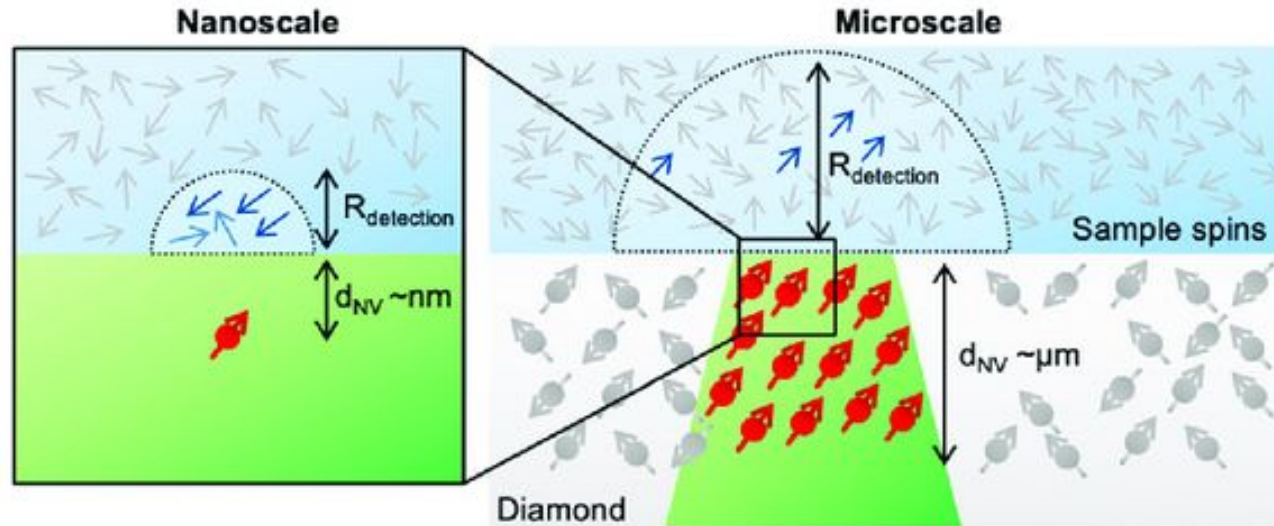
Figure from Bucher et al.

Quantum Microscope

Goals

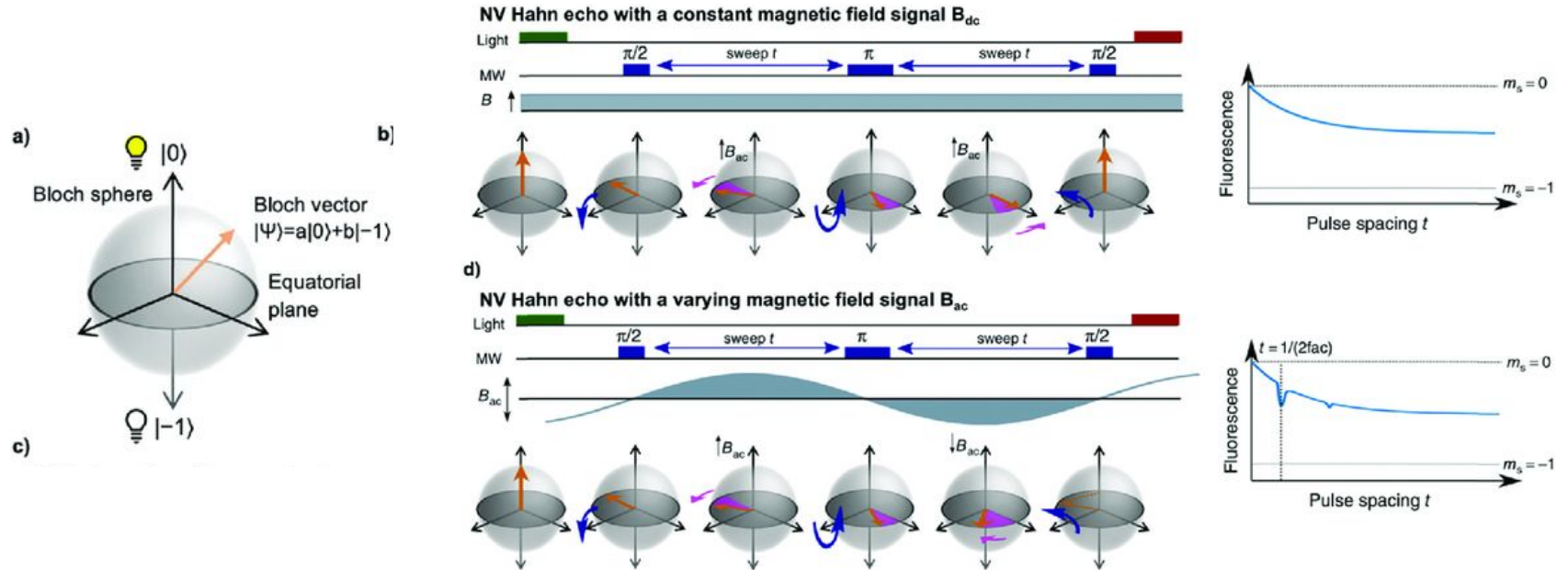
- Optically detected nuclear magnetic resonance (OD-NMR) with smaller sample volumes - nanometer scale!
- Analysis of geochemical samples like confined water

Nuclear magnetic resonance (NMR)

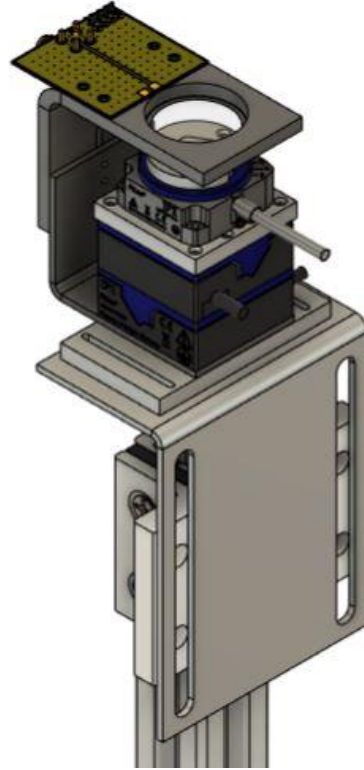
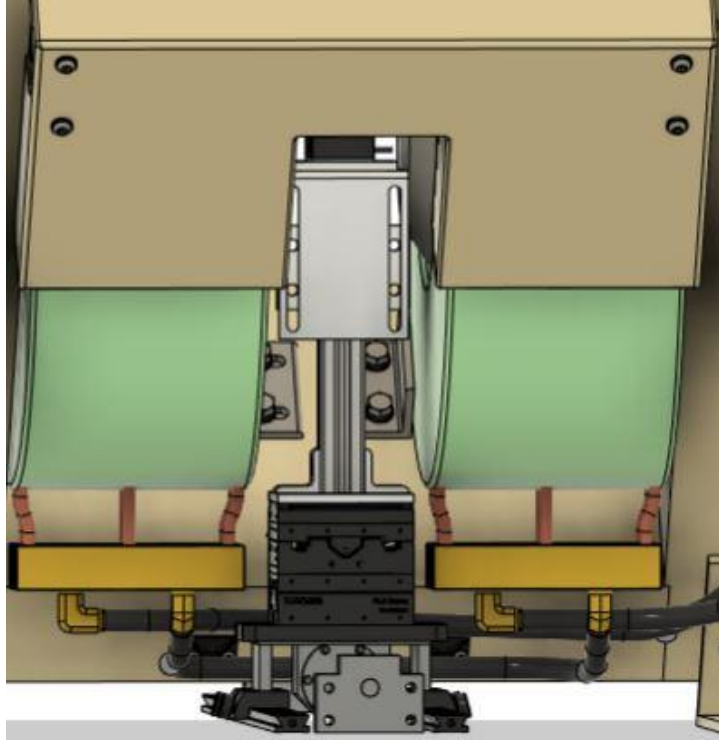


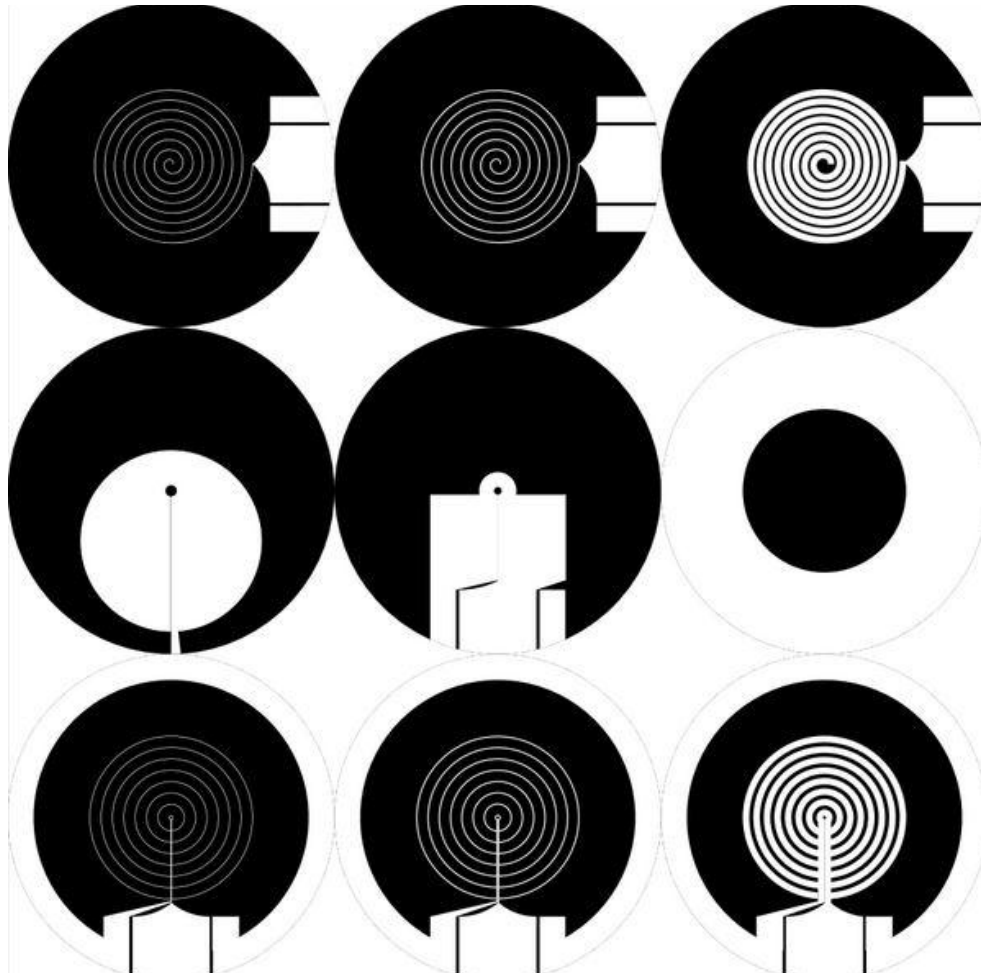
Adapted from Allert et al., 2022

Resonant frequencies $\leftrightarrow$ specific materials



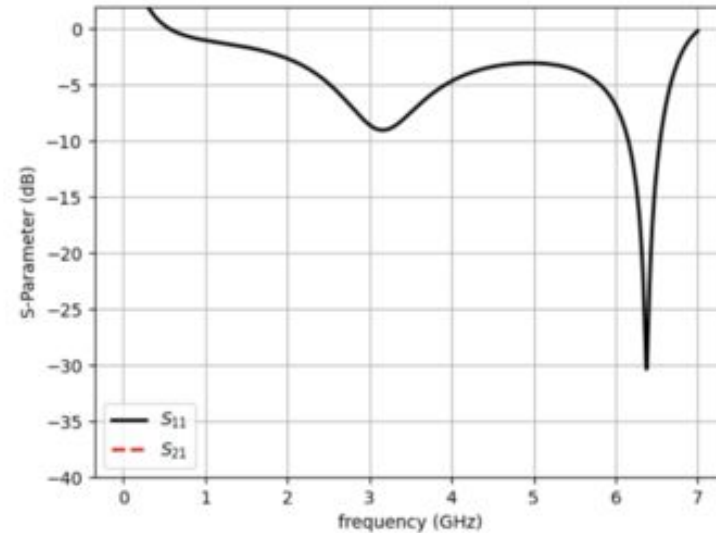
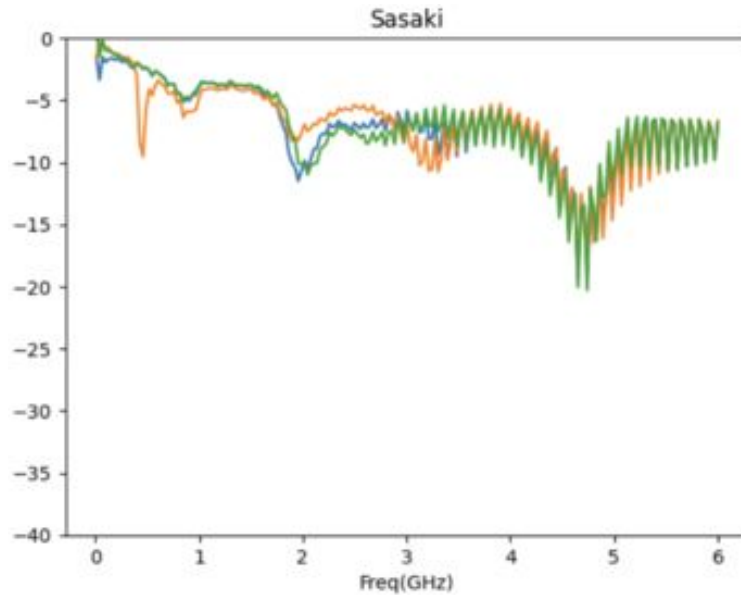
Adapted from Allert et al., 2022



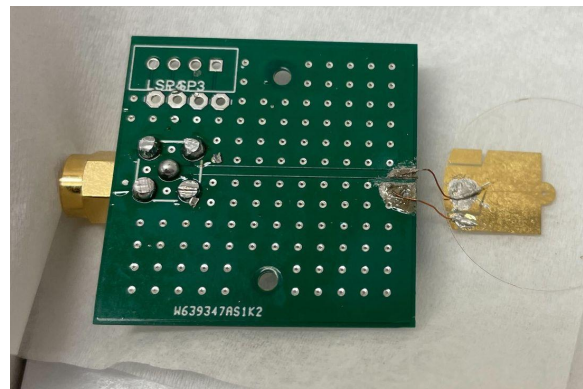
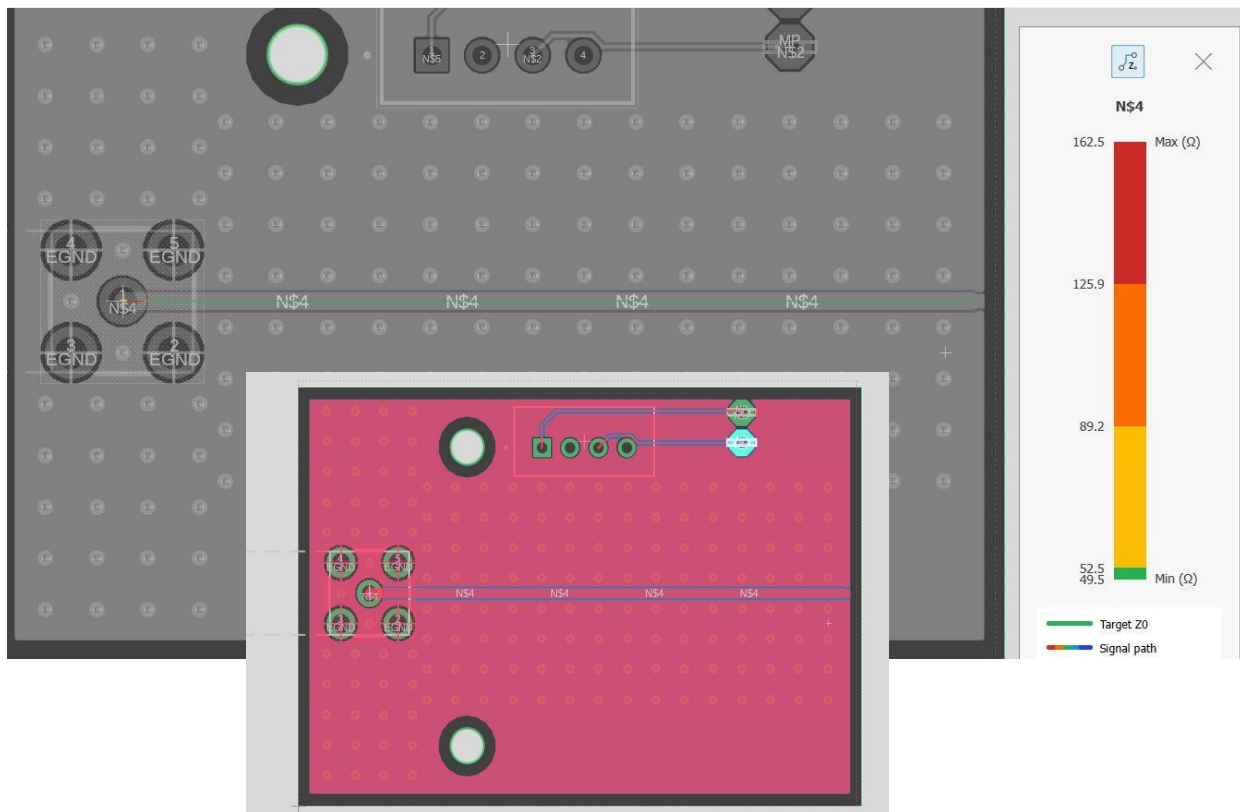


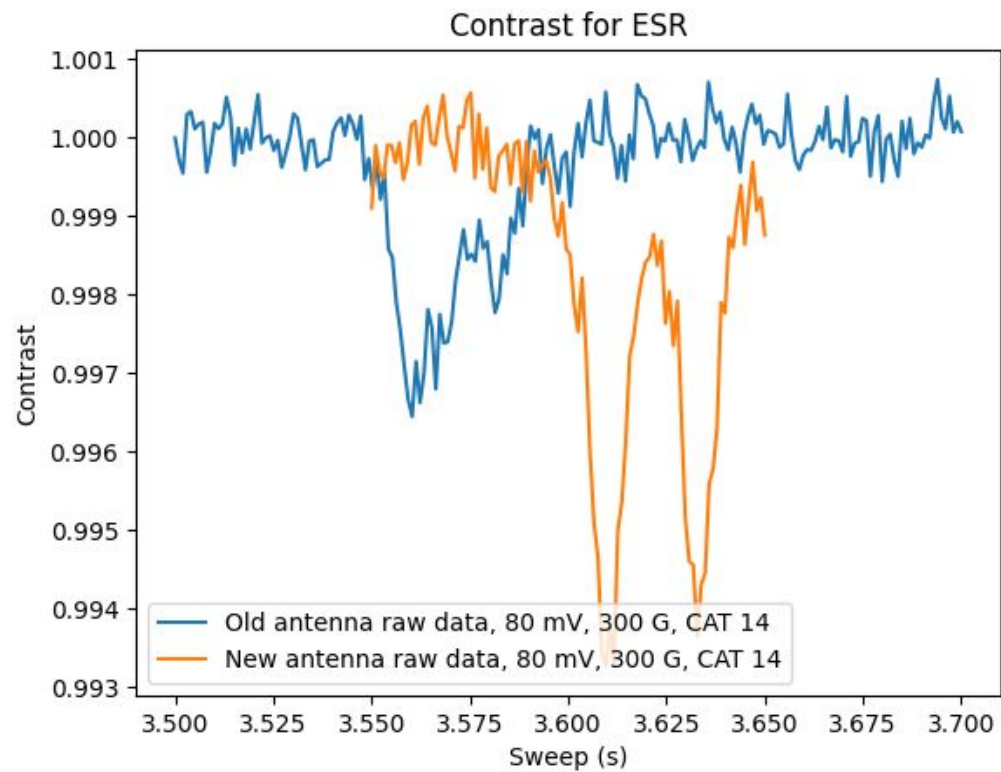
- Mask for antenna manufacture - white is gold, black is glass cover slip/no gold
- Top row: spiral antennas, thickness 100 μm to 500 μm
- Middle row: loop antenna from Sasaki et al., omega antenna from Opaluch et al., ground ring
- Bottom row: new design, concentric antennas with thickness from 100 μm to 500 μm

Simulations (in progress)

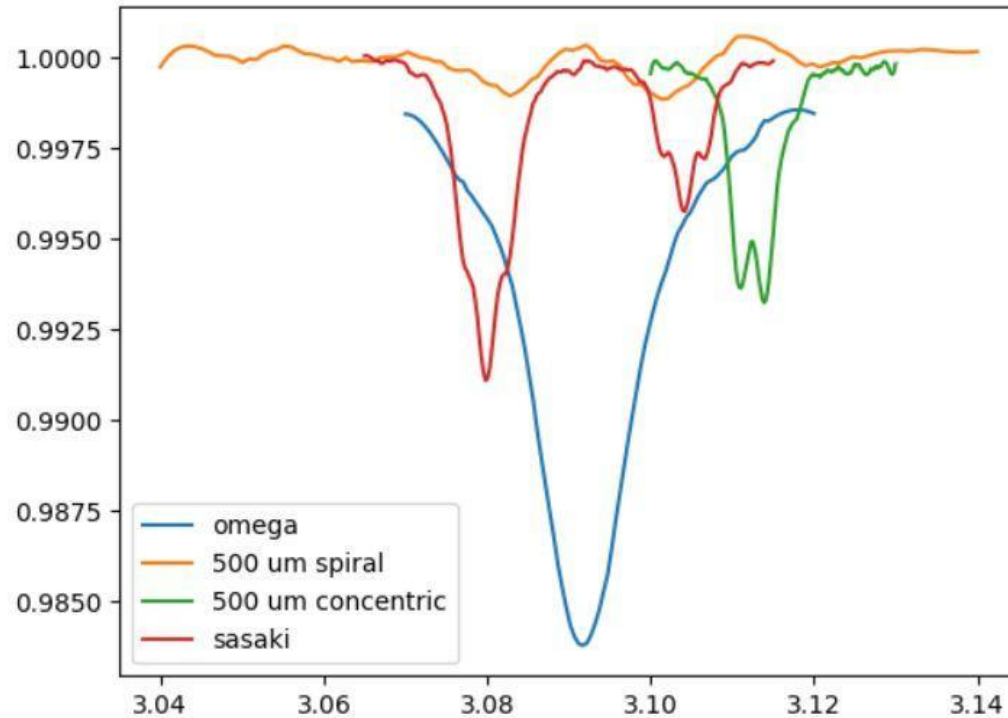


Simulation done assuming PEC - shift in higher peak because of this

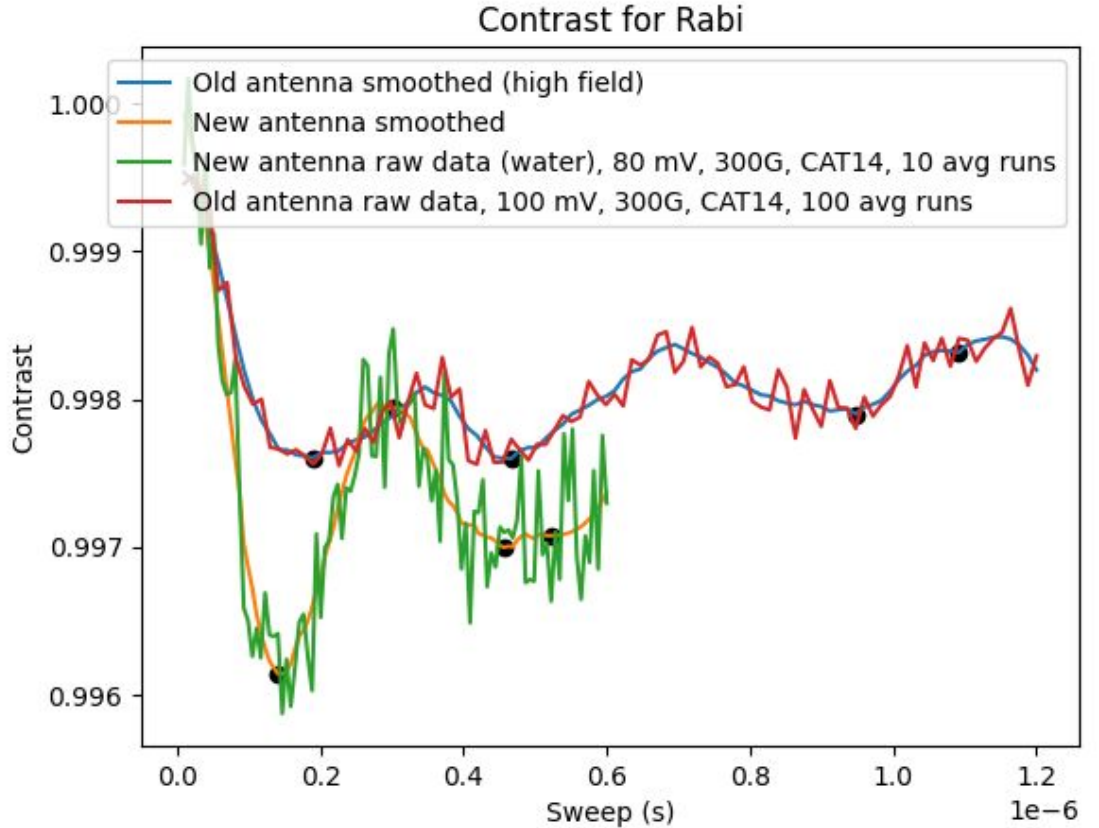




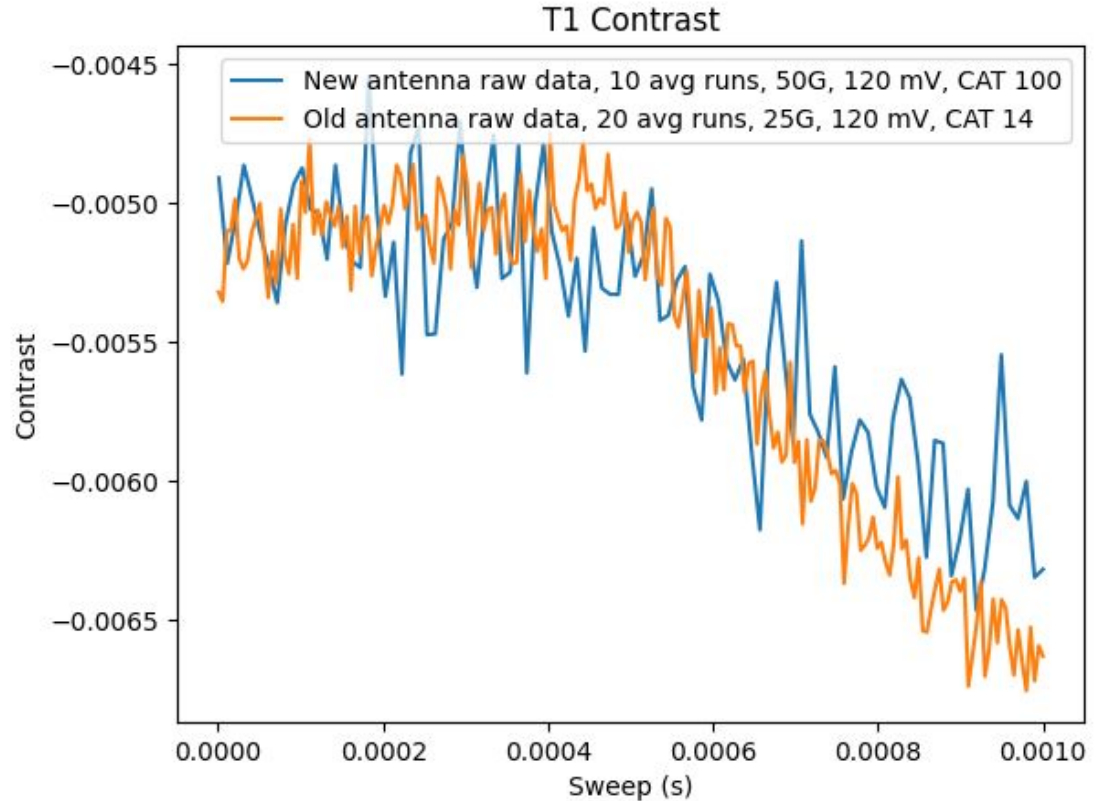
Contrast for ESR - antenna comparison



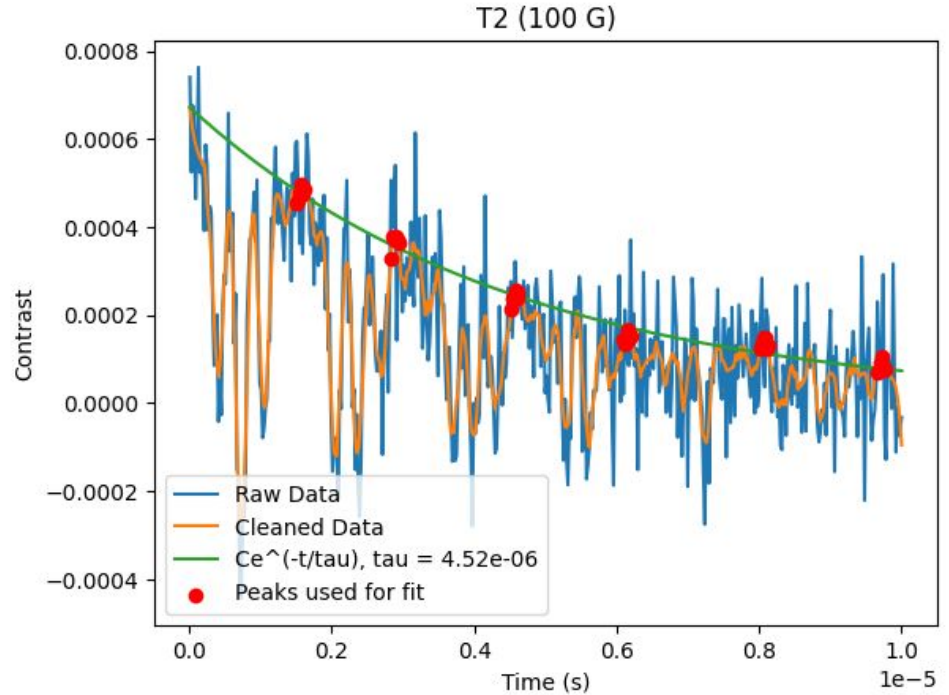
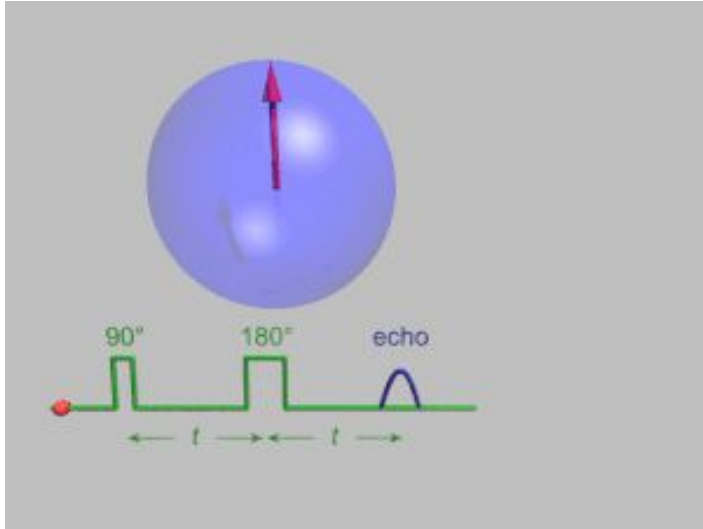
- τ_{π} for new antenna (water rabi): $1.41\text{e-}07$
- τ_{π} for old antenna: $1.90\text{e-}07$



- T1: measures spin-lattice relaxation - relaxation of spin z component back to equilibrium
- Comparable result even with less averaging runs and higher magnetic field



Example pulse train: T2



Animation from Wikipedia of Hahn echo pulse train. T2 measures spin-spin relaxation.

NMR spectra

Magnetometer

Goals

- More sensitive than best (30 pT/ $\sqrt{\text{Hz}}$; Graham et al.)
- Portable, resilient design - want to mount on drone
- Use to detect unexploded ordnance

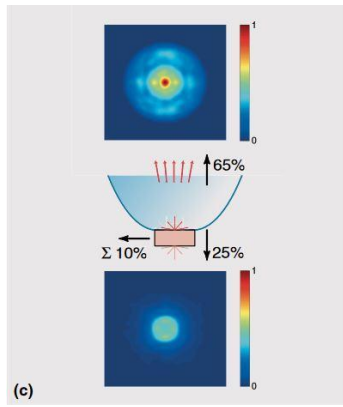
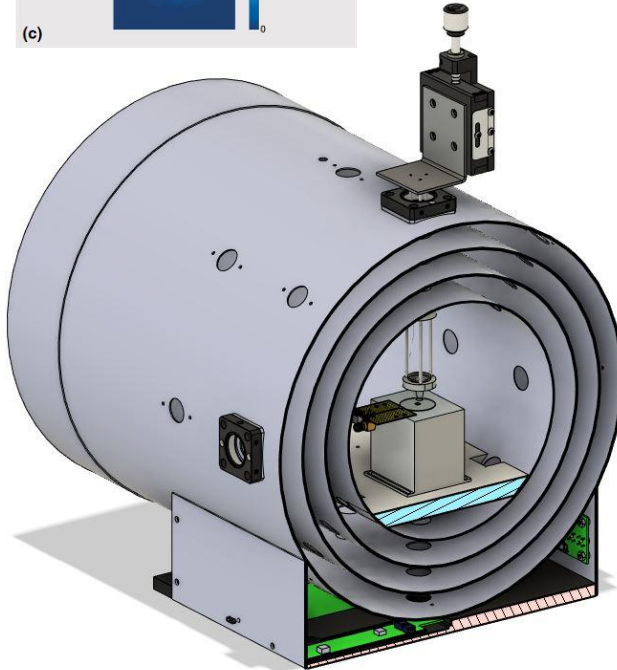


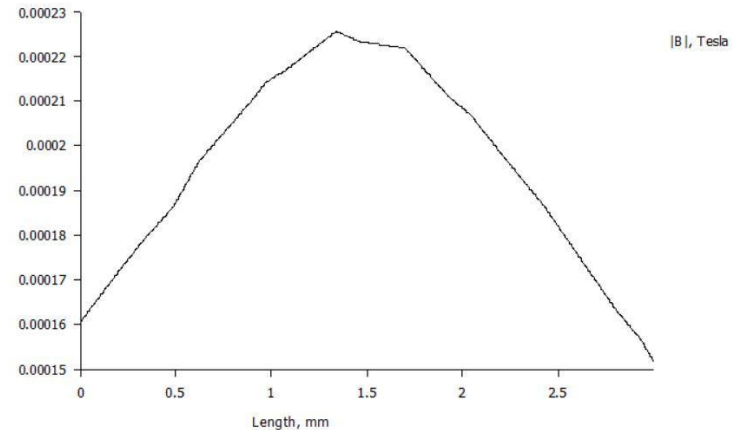
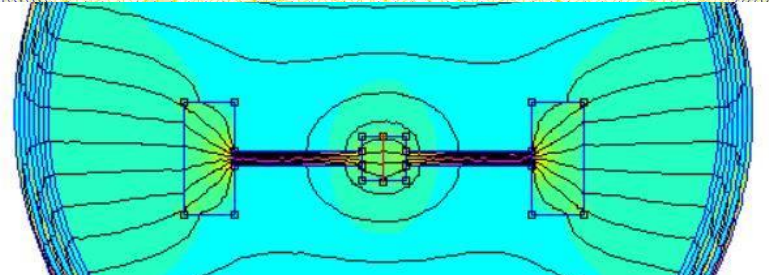
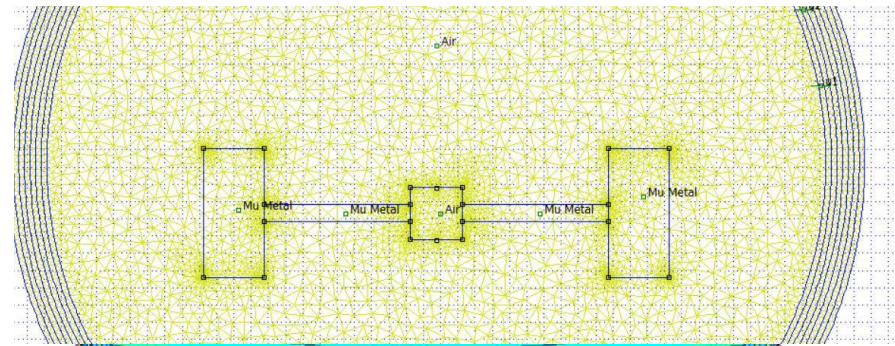
Figure from Wolf et al., 2014

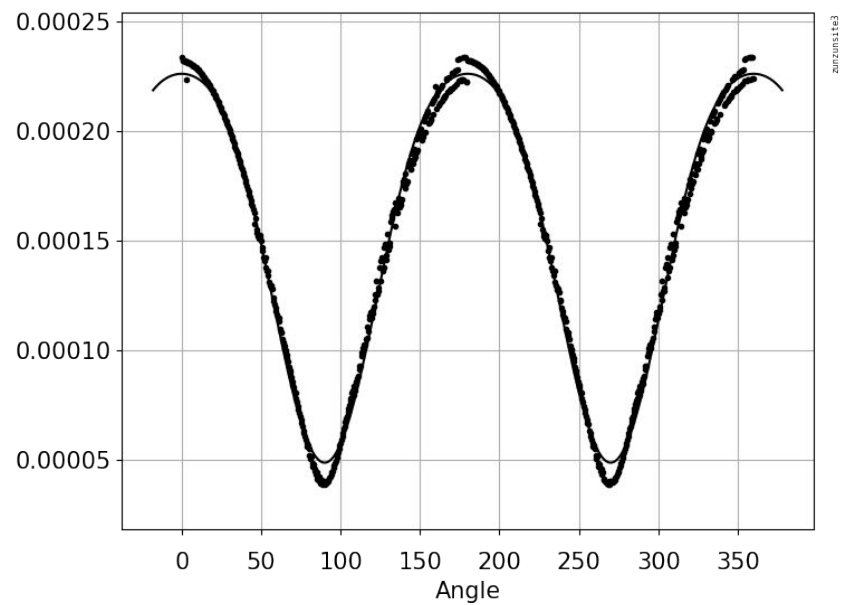
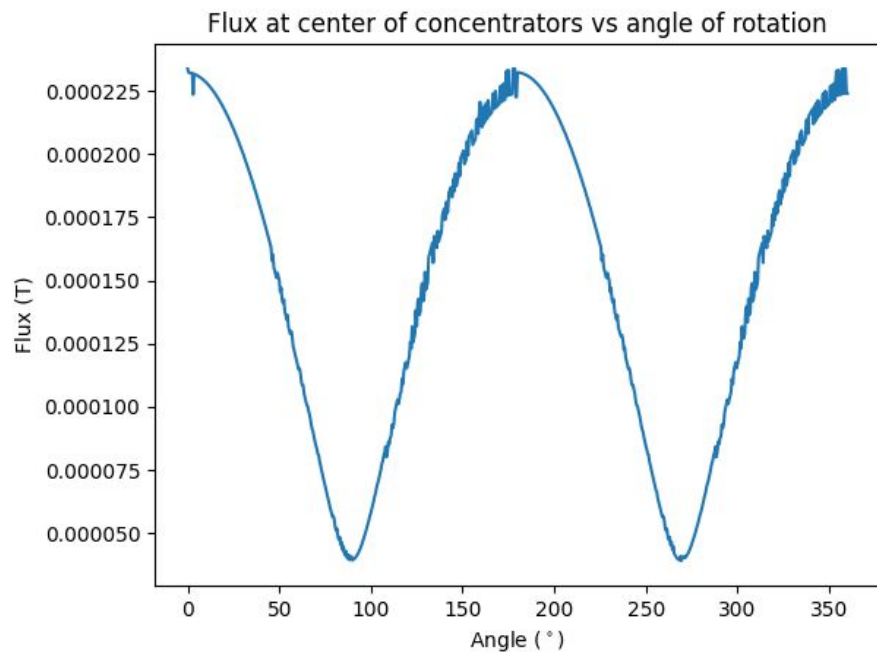


- Magnetic field shielding at right; most of mounting scheme to fit in this
- Parabolic concentrator to collect as much fluorescence as possible
- Antenna design to optimize delivery of microwave pulses
- Signal sensor head and reference sensor head to subtract out background
- Insulation to reduce thermal noise from laser

Flux concentrators

- Simulation done with constant magnetic field of $6.3 \times 10^{-5} \text{ T}$ (Earth's is $5 \times 10^{-5} \text{ T}$) for convenience purposes
- T design based on literature
- Did grid search sweeping width and length of bar and end cap (four parameters)
- Found dimensions for which highest flux concentration occurred: got field in center that was $\sim 3.7\times$ larger, doing further refinement





Especially useful for vector magnetometry

Questions?

Dependence of Flux Concentration on Field Strength

